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To cite this article: Laura Dunne, Aoibheann Brennan-Wilson, Leonardo F. Soares, Dulce Martins Da Silva, Antero Benedito Da Silva, Nicole Gleghorne, Sarah J. Miller & Paul Connolly (2025) Exploring Developmental Outcomes for Children Attending Community-based Pre-Schools in the Ermera and Viqueque Municipalities of Timor-Leste, *Child Care in Practice*, 31:4, 655-664, DOI: [10.1080/13575279.2025.2449852](https://doi.org/10.1080/13575279.2025.2449852)

To link to this article: <https://doi.org/10.1080/13575279.2025.2449852>



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Published online: 03 Feb 2025.



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Exploring Developmental Outcomes for Children Attending Community-based Pre-Schools in the Ermera and Viqueque Municipalities of Timor-Leste

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ABSTRACT

There is a substantial body of evidence which indicates that early childhood programmes have a positive effect on children's school readiness, academic achievement, and social and emotional development. Indeed, these benefits can last into adulthood. Based on this evidence base, many countries that strive to build their early childhood education offer to enhance later chances of prosperity. LINKS is an NIHR-funded study, carried out by a team from Queen's University Belfast and Ulster University, with support from UNICEF. The project aims to explore the impact of early childhood development programmes on social cohesion/peacebuilding in six conflict-affected low- and middle-income countries (LMICs); Timor-Leste, Viet Nam, Mali, Egypt, Kyrgyzstan and Tajikistan. This paper will focus on research carried out as part of the LINKS study in Timor-Leste which aimed to evaluate the impact of UNICEF community-based alternative pre-schools (CBAPs) in Timor-Leste. This research, carried out in 2020, employed a quasi-experimental pre- post-test design with a public pre-school control group. A total of 607 children were recruited to participate from 33 pre-schools: 508 completed post-tests after nine months of pre-school exposure to the CBAP programme. The East-Asia Pacific Early Childhood Development Scales (EAP-ECDS) were used to assess the development of children in the current study. Results of the post-test suggest that despite differences in time children spend in pre-school and infrastructure between the school types, attendance at CBAPs appears to close the developmental gap seen at the pre-test (baseline) between children in the two groups. The findings underline the importance of pre-school access and early learning experiences to the development of children at the highest risk of not reaching their developmental potential. The results point to the need to scale the intervention and to carry out an experimental effectiveness evaluation of this promising programme.

ARTICLE HISTORY

Received 5 October 2023
Accepted 2 January 2025

KEYWORDS

Early childhood education;
child development; low- and
middle-income countries

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Introduction

It has been estimated that approximately 250 million children (43%) younger than 5 years of age in low-income and middle-income countries (LMICs) are at risk of not reaching their developmental potential (Black et al., 2017). There is a substantial body of evidence which indicates that early childhood development (ECD) programmes have a positive short-term effect on children's IQ, school readiness and academic achievement, and social and emotional development (Anderson et al., 2003; Black et al., 2017; Britto et al., 2017; Burger, 2010; Camilli et al., 2010; Richter et al., 2017). Additionally, this research indicates that investment in ECD can have later holistic benefits for children in terms of improved health, educational outcomes and reductions in inequality and poverty (Heckman, 2012). This links directly to the planned achievement of "quality early childhood development, care and education" for all children outlined in the 2015–2030 Sustainable Development Goals (SDGs), by promoting healthy development, reducing inequities and strengthening coordination across various sectors such as education, poverty, climate change, and peace and justice (Carlsen & Bruggemann, 2022; Richter et al., 2017; United Nations, 2015; 2017).

Timor-Leste has a population of 1.3 million, with young people aged 0–17 accounting for 43% of this number. Almost 48% of children live under the national poverty line. Despite significant progress in recent years, the country has one of the highest under-5 mortality rates in the East-Asia Pacific region (41 per 1000 live births) and children in Timor-Leste are also at risk of poor development with high levels of stunting (47%) and wasting (8.6%) (Government of Timor-Leste and UNICEF, 2020). Currently, 9 years of basic education, six at primary level and three at secondary level, are free and compulsory. There is no such mandatory provision at pre-school level. Pre-school enrolment rates remain low (25%) and around 19% children repeat grade one (Government of Timor-Leste and UNICEF, 2020). Pre-school attendance is often further hampered for children living in rural areas for whom access is a barrier to attendance. However, recent research indicates that participation in early childhood education (ECE) is associated with improved cognitive outcomes for young children in LMICs in the East-Asia Pacific region (Rao et al., 2019).

The Ministry of Education, Youth and Sport (MoEYS) has committed to ensure that educational services benefit all Timorese children. Since 2015, UNICEF has supported the implementation of community-based alternative pre-schools with a focus on expanding access to pre-school education in Timor-Leste. The community-based alternative pre-schools (CBAPs) are located in the Ermera and Viqueque municipalities, in aldeias (small villages) where there is no existing pre-school provision. The pre-schools started full operation in January 2016, and in 2019, 3581 children were enrolled in CBAPs. This represents around 46% of pre-school-going children in the target municipalities of Ermera and Viqueque, and about 3% of all children in pre-schools in Timor-Leste overall). In addition to the pre-school curriculum, the programme includes a parenting education and support component (Hametin Familia) aimed at improving parenting practice to support improved developmental outcomes for children. This includes modules on positive discipline, nutrition, and hygiene.

The local communities play an important role in running the CBAPs along with the School Management Committee which is made up of village leaders who oversee the

running of the pre-schools. The engagement of aldeia officials, and village elders is seen as key in terms of mobilising community engagement and support. The pre-schools are run by facilitators who are volunteers from the community and each pre-school has two to three facilitators. Facilitators are nominated by the community and receive training (orientation and refresher sessions, available on demand to revisit aspects of training if required) to enable them to deliver the pre-school curriculum and facilitate parenting education sessions. Classes are undertaken in either the community centre or somebody's home and lessons are based on the Ministry of Education Pre-School Curriculum. These are not designed as educational settings and their learning can be interrupted in favour of other community businesses. Children attend CBAPs for 7.5 h per week (2.5 h per day, 3 days a week). In contrast, public pre-schools are located in designated government-constructed school buildings, often attached to primary schools, and children attend for 20 h each week (4 h per day, 5 days a week). This gives them a more formal status, and means they are seen as being dedicated to educational activities.

Methodology

This study employed a quasi-experimental pre- post-test design with control group to evaluate Community-Based Alternative Pre-schools in the Ermera and Viqueque municipalities of Timor-Leste. This study is part of the NIHR LINKS study, an international research network, in partnership with UNICEF, that is committed to supporting the development and evaluation of early childhood development (ECD) programmes in six low- and middle-income countries impacted by conflict.

Participants

A total of 607 children were recruited from 33 pre-schools in the Ermera and Viqueque municipalities of Timor-Leste (see [Table 1](#)). All the children were from the same class groups within the same schools. The mean age for the sample at baseline was 4.27 (SD = 0.97). The mean age for the children from the Community-Based pre-schools at baseline was 4.19 (SD = 1.02) and 4.42 (SD = 0.83) for the children from the Public pre-schools. The sample included more girls ($n = 319$) than boys ($n = 287$) but the gender balance in each pre-school group was very similar.

There were notable differences between the Community-Based and Public pre-school groups in terms of the children's first language. Tetum is one of two official languages in Timor-Leste, and while the majority of children in the Public pre-school group listed this as their first language, this was only the case for 15.1% of the Community pre-school group. The majority of children (39.4%) in the Community pre-school group listed Makasae as their first language compared with 13.4% of the Public pre-school group. The sample included similar numbers of children from the Ermera (48.0%) and Viqueque (52.0%) municipalities and this breakdown was reflected in the pre-school groups also.

Measure

The East-Asia Pacific Early Childhood Development Scales (EAP-ECDS; Rao et al., 2019) were used to assess the development of all children in the current study. It was developed

and validated as a regional measure in six countries (Cambodia, China, Mongolia, Papua New Guinea, Timor-Leste and Vanuatu) with a total sample of 8439 children (Britto et al., 2017). The measure consists of seven sub-scales, the scores of which are combined to give a total score:

1. Cognitive Development;
2. Socio-emotional Development;
3. Motor Development;
4. Language and Emergent Literacy;
5. Health, Hygiene, and Safety;
6. Cultural Knowledge and Participation; and
7. Approaches to Learning

Data collection

Data collection was conducted by the research team from the Peace Centre at the Universidade Nasional Timor Lorosa'e (UNTL). Training in the administration and scoring of the EAP-ECDS measure was provided by the developers of the measure from Hong Kong University, over two days in November 2019. The training took place in UNTL and included the in-country research team and fifteen local fieldworkers, recruited by the UNTL team. The fieldworkers were recruited by the UNTL team through a selection process where they applied and had to demonstrate their capacity and experience in data collection, engaging with communities and the ability to communicate using local languages. During the recruitment process, the team was keen to recruit suitably qualified people from the municipalities of Ermera and Viqueque as they could speak the local languages of these two areas. There are four mother languages, Makasae, Naueti, Tetun Terik and Cairui, in Viqueque and two mother languages, Mambae and Kemak, in Ermera. Despite the diverse ethno-linguistic context, most of the parents and children were actually able to speak Tetun, the national language and language of unification in Timor L'este. In the few cases where children did not speak Tetun, the fieldworkers used local language instead.

Table 1. Sample characteristics of children recruited from community-based and public pre-schools at baseline.

	Community-based pre-schools	Public pre-schools
Mean age (SD)	4.19 (1.02)	4.42 (0.83)
Gender		
Male	195 (48.3%)	92 (45.3%)
Female	209 (51.7%)	111 (54.7%)
First language		
Kairui	0 (0.0%)	1 (0.5%)
Kemak	91 (22.5%)	0 (0.0%)
Makasae	159 (39.4%)	27 (13.3%)
Mambae	74 (18.3%)	68 (33.5%)
Naueti	19 (4.7%)	0 (0.0%)
Tetum	61 (15.1%)	73 (36.0%)
Tetum Terik	0 (0.0%)	34 (16.8%)
Location		
Ermera	194 (48.0%)	98 (48.3%)
Viqueque	210 (52.0%)	105 (51.7%)
Total	404 (66.3%)	203 (33.7%)

Pre-test data was collected in February 2020 at the beginning of the school year; post-test data was collected in November 2020, towards the end of the school year. All schools closed for a period of two months between the end of March 2020 and the end of May 2020, due to the Covid-19 pandemic.

Data collection took place in pre-school settings in each of the target municipalities. Public pre-schools were generally located in designated buildings while community pre-schools were typically located in community houses, or community centres. Fieldworkers visited each setting and completed surveys on a one-to-one basis with the children. All responses were recorded and entered on tablet devices using KoboToolbox software to be stored in a repository managed by the Harvard Humanitarian Institute. The data from the EAP-ECDS was coded and scored according to the scoring procedure outlined in the test manual (Rao et al., 2019). Data were analysed using Stata 17.

Ethical approval

Ethical approval to conduct the study was granted by the Research and Ethics Committee at the School of Social Sciences, Education and Social Work at Queen's University Belfast (Ref: 027_1920). The QUB team funded and directed the research. Informed consent was obtained for all participant groups in the study by the UNTL research team. Parents were asked to provide informed consent for their children to participate. Ongoing verbal assent to participate was also obtained from children during data collection. The children were given the right to refuse to take part in the study and to withdraw at any time throughout. This was explained to them by the UNTL research team, which included local fieldworkers who could communicate in the local language. Privacy and confidentiality were maintained throughout the study.

Results

The performance of each pre-school group on the EAP-ECDS (composite score and sub-scales) was analysed using multi-level models (pupils nested within schools) at baseline (pre-test) to determine the comparability of the two groups (Table 2). The child's score was used as the dependent variable and a dummy variable for pre-school type was included as the independent (predictor) variable. Dummy variables for location and gender were also included as control variables, along with age (standardised). As outlined in Table 2, children in Public Pre-schools performed significantly better than their Community pre-school counterparts at baseline on all elements of EAP-ECDS (composite score and sub-scales), with the exception of the Approaches to Learning sub-scale.

A total of 508 ($n = 153$ control; $n = 355$ intervention) children completed post-test, giving a retention rate of 83.7% (control 75.4%; intervention 87.9%). The final analysis population included only those children with both baseline and post-test data. Two-level multi-level models were used to compare the performance of the two pre-school groups on the EAP-ECDS (composite and sub-scales) at post-test (9-month follow-up), controlling for pre-test scores, age, gender and location, with pre-school as the level 2 covariate.

At post-test, the children in Public pre-schools continued to score higher than their Community pre-school counterparts on the EAP-ECDS (composite score and sub-scales) but this difference only reached significance for the Motor Development (p

Table 2. Comparability between community and public pre-school groups at pre-test (baseline).

Outcome	Community PS		Public PS		Effect size (95% CI)	Sig.
	Adjusted ^a Mean (SD)	N	Adjusted ^a Mean (SD)	N		
Cognitive development	6.17 (1.68)	404	8.91 (1.51)	203	−0.25 (−0.42 to −0.08)	.011
Socio-emotional development	4.64 (1.30)	402	8.54 (1.12)	201	−0.35 (−0.52 to −0.18)	<.001
Motor development	4.38 (0.71)	377	5.50 (0.58)	197	−0.31 (−0.48 to −0.14)	.001
Language and literacy development	4.05 (1.77)	391	7.26 (1.66)	197	−0.30 (−0.47 to −0.13)	.001
Health, hygiene & safety	3.48 (1.39)	375	6.78 (1.18)	195	−0.51 (−0.68 to −0.33)	<.001
Cultural, knowledge & participation	2.82 (1.01)	371	5.30 (0.89)	191	−0.50 (−0.68 to −0.32)	<.001
Approaches to learning	1.87 (0.41)	390	2.67 (0.34)	190	−0.22 (−0.39 to −0.04)	.026
EAP-ECDS total score	26.74 (8.26)	404	43.57 (7.25)	203	−0.36 (−0.53 to −0.19)	<.001

^aAdjusted for age, gender and location.

= .025) and Cultural Knowledge & Participation sub-scales ($p = .051$; see Table 3). There was no significant difference between the two groups on children's Cognitive Development, Socio-Emotional Development, Language & Literacy, Health, Safety & Hygiene, or Approaches to Learning sub-scale scores, reflecting a shift from baseline scores (see Figure 1).

Discussion

This work highlights the importance of pre-school access and early learning experiences to the development of children at highest risk of not reaching potential. Results from pre- and post-test EAP-ECDS suggest that despite differences in amount of exposure to the programme and setting infrastructure, children in CBAPs appear to close the developmental gap between themselves and their Public pre-school counterparts. Table 2 shows that the Public pre-school group significantly out-perform the CBAP group on all domains of the EAP-ECDS at baseline. By post-test (Table 3), it can be seen that this gap has narrowed considerably, with the Public pre-school group only significantly scoring higher than the CBAP group on Motor Development. This is illustrated in Figure 1.

It is important to remember that children in CBAPs are typically only exposed to seven and a half hours of class time per week, spread over three days, whereas children attending Public pre-schools have much greater exposure to education, around twenty

Table 3. Adjusted post-test scores for the community and public pre-school groups.

Outcome	Community PS		Public PS		Effect size (95% CI)	Sig.
	Adjusted ^a Mean (SD)	N	Adjusted ^a Mean (SD)	N		
Cognitive development	10.61 (14.31)	354	11.97 (13.67)	153	−0.10 (−0.29 to 0.09)	.243
Socio-emotional development	8.38 (12.03)	347	9.92 (10.96)	151	−0.32 (−0.32 to 0.06)	.139
Motor development	5.47 (3.09)	347	6.13 (2.69)	150	−0.22 (−0.41 to −0.03)	.025
Language and literacy development	7.52 (13.28)	344	8.73 (13.19)	152	−0.09 (−0.28 to 0.10)	.286
Health, hygiene & safety	6.77 (8.15)	332	7.63 (6.89)	149	−0.11 (−0.30−0.08)	.225
Cultural, knowledge & participation	4.45 (7.06)	332	5.70 (6.15)	148	−0.18 (−0.38 to 0.01)	.051
Approaches to learning	3.41 (3.58)	330	3.92 (3.31)	149	−0.15 (−0.34 to 0.05)	.103
EAP-ECDS total score	45.78 (24.55)	354	51.58 (23.79)	153	−0.24 (−0.43 to −0.05)	.209

^aAdjusted for pre-test score, age, gender, location.

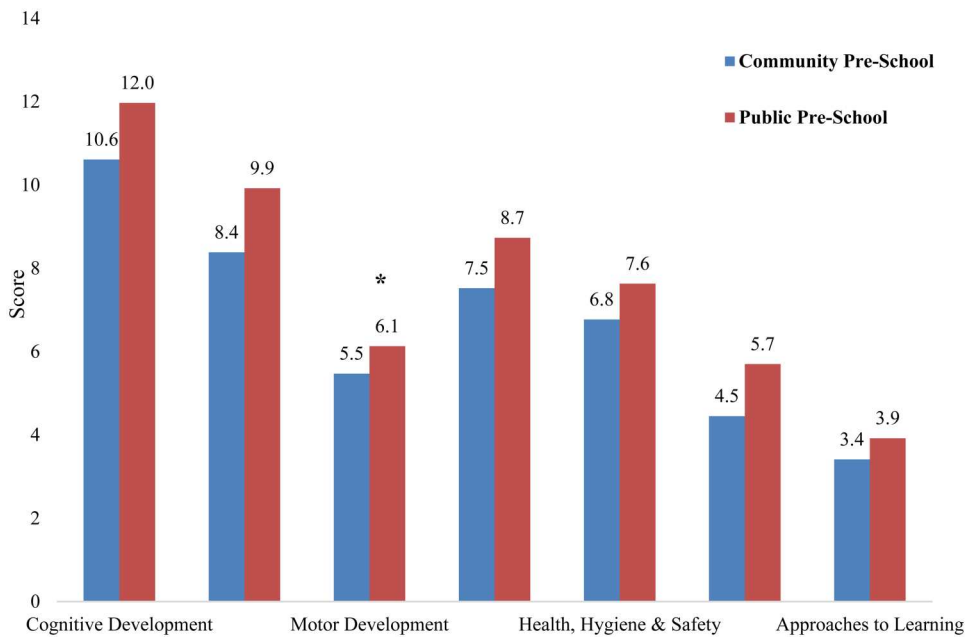


Figure 1. EAP-ECDS sub-scale scores at post-test for children attending CBAPs or public pre-schools. * $p < 0.05$.

hours of tuition a week, spread over five days. The context where learning takes place is also important. Learning in CBAPS takes place in less formal settings than Public pre-schools, usually based in a community centre or house, where interruptions in favour of everyday activities or community business are common. On the other hand, Public pre-schools are typically based in purpose built, government-constructed buildings, often attached to primary schools, perhaps lending them a more formal or “official” status.

Given the fewer hours of exposure to learning and the more informal setting, it is therefore notable that such positive change was observed in the CBAP group.

While there are still substantial numbers of children who cannot access pre-school provision in Timor-Leste, the government has committed significant resources to ensuring higher levels of access (Government of Timor-Leste and UNICEF, 2020). Not only would it be useful to expand and test the CBAP model from “small to bigger” it would be important to also adapt it from “bigger to better” (Yoshikawa et al., 2018). Some of the areas this might be developed in include resourcing dedicated infrastructure and broad evidence-based curricula. In terms of scale, there are many factors at a systems level that need to be considered before interventions can be rolled out at national, sub-national and local levels. These require input and buy in from a range of multi-sectoral partners ranging from government to community representatives (Yoshikawa et al., 2018).

One limitation of this work was that it was a quasi-experimental design. This was necessary as the settings had already been allocated by UNICEF before the study design was agreed, making randomisation impossible. In order to enhance the evidence

around CBAPs, it would be important to evaluate effectiveness in the most rigorous way, using a randomised controlled trial design. In this way, findings could be more secure, and differences between groups attributable to the treatment group alone (Connolly et al., 2017). Although fieldworkers were able to adapt to speaking languages most familiar to the children, some items in the EAP-ECDS were unfamiliar to the children, such as pictures of an apple or a train. Children often tried to appropriate something familiar to them, calling an “apple” a “tomato” or calling a “train” a “house”. While earlier research reports the EAP-ECDS as a reliable measure for use in the region, more work is needed to ensure that adequate and appropriate cultural adaptation of the intervention and measures used, aligned with training given to fieldworkers and taking culture and context into account (Giordano & Ungar, 2021; Perera et al., 2020). It is common, for example, for children to use local materials and traditional games to learn to count.

Overall, this work demonstrates promise in terms of potential quality and equitable pre-school provision for children in Timor-Leste and underlines the directive by Lu et al. (2020) that “efforts of reducing inequalities must focus on the poorest families and those living in rural areas in the poorest countries”.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Institute for Health and Care Research (NIHR) [grant number 16/137/85].

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Sarah J. Miller’s research revolves around three inter-related themes: Social-emotional development, academic attainment and programme evaluation. She has designed, conducted and published a number of large-scale studies, which have included: simple and cluster randomised controlled trials, systematic reviews, quasi-experimental evaluations and cross-sectional surveys.

Paul Connolly, is Pro-Vice-Chancellor and Executive Dean of the Faculty of Arts, Humanities and Social Sciences at Ulster University, Northern Ireland. He is also a professor of Education and internationally known for his research on early childhood, diversity and inclusion.

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